24ME181
ENGINEERING GRAPHICS

Course Category:	Engineering Sciences (ES)	**Credits:**	**2.5**
Course Type:	Tutorial & Practice	**Lecture-Tutorial-Practice:**	**0-1-3**
Pre-requisites:	**Nil**	**Continuous Assessment:** **Summative Assessment:** **Total Marks:**	**60** **40** **100**

Course Description:

This course introduces students to the principles and practices of engineering graphics using Computer- Aided Design (CAD) tools. This course covers Plane and Descriptive geometry, orthographic and isometric projections. Students will learn to create, modify, and analyze 2D and 3D drawings, focusing on applications in engineering.

Course Objectives:

- To familiarize students with CAD software and its application in technical drawing.
- To make students proficient in creating and editing 2D and 3D engineering drawings.
- To teach the drawing of plane curves and their applications
- To explain the projection of points, lines, Planes, and solids.
- To enable students to visualize and represent 3D objects through orthographic projections, isometric views and development of surfaces.
- To cultivate skills in preparing engineering drawings following standard conventions and practices.

Course Outcomes:

At the end of the course, the students will be able to

CO	Course Outcomes	BTL
CO1	Construct plane curves and identify their applications.	**K3**
CO2	Draw orthographic projections of points, lines and planes	**K3**
CO3	Develop 2D drawings of solids and surfaces.	**K3**
CO4	Demonstrate proficiency in creating isometric projections.	**K4**
CO5	Identify the 3D objects through orthographic projections.	**K4**

Course articulation matrix

	POs											PSOs	
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**
CO1	**2**		**2**		**2**				**2**				**1**
CO2	**2**		**2**		**2**				**2**				**1**
CO3	**2**		**2**		**2**				**2**				**1**
CO4	**2**		**2**		**2**				**2**				**1**
CO5	**2**		**2**		**2**				**2**				**1**

(1-Low, 2 -Medium, 3–High)

Course Content

Unit-I

Plane Curves

- Construction of **Ellipse, Parabola and Hyperbola** (General method only)
- Construction of **cycloids & Involutes.**

Unit-II

Orthographic Projections:

- Projections of **Points.**
- Projections of **Lines** (First Angle Projection only)
- Projections of **Plane regular geometric figures**. (First Angle Projection only)

Unit-III

Projection of Solids and Development of Surfaces

- Projections of simple solids such as Cubes, Prisms, Pyramids, Cylinders and Cones with varying positions (Limited to Solid Inclined to one of the Reference planes)
- Development of surfaces of Right Regular Solids of Prism, Pyramid, Cylinder and Cone.

Unit-IV

Conversion to orthographic Projection

- Orthographic projection conventions.
- Conversion of Isometric views into orthographic projections

Unit-V

Conversion to Isometric Projection

- Isometric Projection Principles, Isometric Views of Objects.
- Conversion of Orthographic Projections into Isometric Views

Text Books:

1.Kulkarni, D. M., Rastogi, A. P., & Sarkar, A. K. (2009). Engineering graphics with AutoCAD, (revised ed.). PHI Learning Pvt. Ltd.

2. N.D. Bhatt "Engineering Drawing", Charotar Publishing House, Anand. 53rd Edition – 2019.

Reference Books:

1. Bethine. A. Fishel, Jay D. Helsel, (2014). Engineering Graphics with AutoCAD, Pearson Education.
2. CADFolks, (2023). AutoCAD 2023 for Beginners, CAD Folks Publications.
3. BasanthAgrawal& C M Agrawal," Engineering Drawing", McGraw Hill Education Private Limited, New Delhi.

Web Resources:

[1] https://www.onlinecourses.nptel.ac.in/noc20_me79/preview
[2] https://www.autodesk.com/education
[3] https://www.freecad.org
[4] https://ocw.mit.edu/